Phase 7 plan — Nursing Station (supervisor role) + Patient Character Management + Telemetry/ECG/Intercom

Status: PLAN — not yet started. **Authored:** 2026-05-26 (post-MVP gate clarification). **Methodology:** Mirrors the v6 Development Plan's M-style modules. Each section below is a self-contained module specification with Purpose, Files Touched, New Files, Acceptance, Blocks/Blocked-by, and Effort Estimate.

0. Why this phase exists — terminology clarification

The MVP build conflated two roles under one name:

1. **Instructor's operator surface.** The browser the instructor sits at to set up encounters, freeze the room, inject scenes, manage simulated patients. This is the **Control Room**. 2. **In-simulation supervisory role** played by a student. A charge nurse / floor supervisor who monitors a roomful of patients remotely — telemetry strips, device status, alarm board, intercom to any bed. This is the **Nursing Station**.

In v7 M5, the instructor surface at `/portal/room` was labeled "Charge-nurse dashboard." That name belongs to the in-sim student role, not the instructor. Phase 7 corrects the name, **extracts patient-character-management features into a dedicated instructor sub-screen**, and **adds the Nursing Station as a new student role** with the telemetry / device / alarm / intercom features that role requires in a real clinical workflow.

Conceptual model after Phase 7

```
INSTRUCTOR (Operator) — /portal/room and children
■■■ Multi-Patient Control (renamed from "Charge-nurse dashboard")
■   ■■■ Overview grid       (existing M5 — kept)
■   ■■■ Wizard               (existing M6 — kept)
■   ■■■ Per-Patient Console  (NEW — M22 extraction)
■       ■■■ Live telemetry preview (M23)
■       ■■■ ECG waveform picker     (M24)
■       ■■■ Device list + state     (uses v6 device subsystem)
■       ■■■ Quick alarm inject       (writes through M26 alarm bus)
■       ■■■ Telemetry overrides      (instructor force-set values)

STUDENT (in-simulation roles)
■■■ Bedside (existing — one student per chat station)
■   ■■■ Chat station       (v3)
■   ■■■ EHR station         (v5)
■   ■■■ Device station      (v6 — pumps/cabinets)
■       + future: call bell, bed alarm, code blue button (M29)
■■■ NEW — Nursing Station role
    ■■■ Multi-patient telemetry view  (M27 page)
    ■■■ Device status per patient      (read-only)
    ■■■ Alarm board                    (M26 alarm bus)
    ■■■ Intercom                       (M28 — voice to any bed)
```

1. Existing modules to modify

Five small touch-ups to land before the new modules. Each is < 0.5 day; all stay v6-compatible.

1.1 Rename "Charge-nurse dashboard" → "Multi-Patient Control"

The label on the instructor surface, in nav, code comments, and docs. Code paths stay at `/portal/room`. Variable names in `control_room.html`/`.js`/`.css` keep `room-*` prefixes — only the human-readable label changes.

Files touched:

• `portal/templates/control_room.html` (title block, `<h1>`)

• `portal/templates/base.html` (nav label: `Room (multi)` → `Multi-Patient Control` or `Patient Management`)

• `portal/static/control_room.{js,css}` (header comments only)

• `portal/server.py` (docstring comments)

• `docs/module_guides/M04_*`, `M05_*`, `M06_*`, `M09_*` (text only)

• `BUILD_STATE.md` (text only)

Acceptance: existing 58 v7 tests still pass; preview shows the new title in the page header and nav.

Effort: 0.25 day.

1.2 Add a "role" field to the Student row

Students playing the Nursing Station role need to be distinguishable from bedside students at the data layer so M27's join flow can route them correctly.

Files touched:

• `portal/ehr_db.py` — schema migration v5 (append-only):

``sql ALTER TABLE student ADD COLUMN role TEXT NOT NULL DEFAULT 'bedside'; ``

• `portal/control_room.py` — `Student` dataclass adds `role: str = "bedside"`

• `portal/ehr_db.py` — CRUD helpers updated to round-trip the new column

Acceptance: migration v5 idempotent + preserves v4 data; existing rosters default to `bedside`; new `register_student(..., role=...)` persists.

Effort: 0.5 day.

1.3 Extend the M5 dashboard with a per-encounter drill-in slot

Currently clicking a card on `/portal/room` navigates to `/portal/control/ops?join=<code>` (the v6 single-encounter ops view). M22 introduces the new Per-Patient Console at `/portal/room/encounter/<id>`; M1.3 only adds the routing target (the page itself is M22). For backwards compat, the legacy ops link remains available as a secondary option.

Files touched:

• `portal/static/control_room.js` — primary card click navigates to

`/portal/room/encounter/<id>`; a "Legacy ops view" link on the card stays as a secondary entry.

Acceptance: click test in preview confirms the new URL; legacy ops link still works.

Effort: 0.25 day.

1.4 Extend the scenes engine to flag alarms

scenes.code.blue already writes a note.save "CODE BLUE" announcement + instructor.trigger marker + (if pump bound) device alarm. M26's alarm bus needs to find these. Add a level: "alarm" tag in the _scene_payload of any alarm-class scene.

Files touched:

• portal/scenes.py — payload tag, no behavior change.

Acceptance: existing M7 tests pass unchanged; new tag is in the payload for code.blue.

Effort: 0.1 day.

1.5 Extend the device subsystem to enumerate device kinds

The Per-Patient Console (M25) lists devices currently bound to an encounter. The device subsystem already supports pump_iv, pump_enteral, and cabinet. M29 will add 4 more kinds; M22 needs the existing list exposed via a stable API.

Files touched:

• portal/devices/registry.py — add a list_kinds() helper.

Acceptance: a unit test confirms the helper returns the known device kinds.

Effort: 0.1 day.

Pre-Phase 7 total: ~1.5 days.

2. New modules

Eight modules, ~17 engineer-days, mirrors the v6 plan's M-shape.

MODULE M22 — Per-Patient Console scaffold (instructor)

Goal: Extract the "manage one simulated patient's state" surface from the M5 overview into a dedicated instructor sub-page at /portal/room/encounter/<encounter_id>. M22 ships the scaffold — the rich features (telemetry preview, ECG picker, alarm inject) land in M23 / M24 / M25.

• **New files:**

• portal/templates/encounter_console.html

• portal/static/encounter_console.{js,css}

• **Files touched:**

• portal/server.py — new GET /portal/room/encounter/{encounter_id}

route returning the scaffold.

• **Acceptance:**

• Route renders 200 with the right encounter id.

• "Back to overview" link returns to /portal/room.

• Browser preview smoke test on a 2-bed quickstart room.

• **Blocks:** M23, M24, M25.

• **Blocked by:** M1.3 (drill-in target).

- **Est. effort:** 1.5 days.

MODULE M23 — Telemetry simulation engine (`portal/telemetry.py`)

Goal: Derive a continuous telemetry snapshot per encounter from its latest `vitals.record` chart event, with optional instructor overrides. No physiological model — just smoothed sampling around the last set vitals (small jitter ±2 BPM HR, ±1 mmHg BP, etc.) so the displayed numbers look live rather than static.

- **New files:**
- `portal/telemetry.py` — `snapshot(encounter_id) -> dict`, `override(encounter_id, key, value)`.
- **Files touched:**
- `portal/ehr_db.py` — schema migration v5 adds `ehr_session.telemetry_overrides_json TEXT NULL`.
- **New routes (in `server.py`):**
- `GET /api/encounter/{id}/telemetry` — poll snapshot.
- `POST /api/encounter/{id}/telemetry/override` — instructor force-set.
- **Acceptance:**
- `tests/v7/test_telemetry_snapshot_uses_latest_vitals.py`
- `tests/v7/test_telemetry_scene_injection_changes_snapshot.py`
- `tests/v7/test_telemetry_overrides_take_precedence.py`
- **Blocks:** M25, M27.
- **Blocked by:** M1.2 (schema v5 housekeeping).
- **Est. effort:** 2 days.

MODULE M24 — ECG waveform library (`portal/ecg.py`)

Goal: A static catalog of named ECG waveforms keyed by rhythm class. Each entry is a parameterized waveform generator that the client renders as an SVG path. Initial catalog:

Waveform id	Class	Common use
`nsr`	Normal	Baseline; default for a stable patient
`sinus_tachy`	Tachy	Septic / dehydrated / anxious patient
`sinus_brady`	Brady	Athlete; vagal; medication effect
`afib`	Irregular	AFib with controlled or RVR
`aflutter`	Regular	Atrial flutter with 2:1 / 4:1
`vtach_mono`	Wide-complex tachy	Monomorphic VT
`vtach_poly`	Wide-complex tachy	Torsades de pointes
`vfib`	Chaotic	Ventricular fibrillation
`asystole`	Flatline	Cardiac arrest
`pea`	Organized w/o pulse	Pulseless electrical activity
`paced`	Paced rhythm	After pacemaker

Each entry includes: rate range (default + scenario-tunable), amplitude, characteristic intervals. The client renders the waveform in a scrolling strip from these parameters; no recordings, no external audio.

• **New files:**

• `portal/ecg.py` — catalog + generator helpers (server-side; the

actual SVG rendering happens client-side). - `portal/static/ecg_strip.js` — renderer (used by M25 and M27).

• **Files touched:**

• `portal/ehr_db.py` — schema migration v5 adds

`ehr_session.ecg_waveform_id TEXT NULL`. - `portal/server.py` — `GET /api/ecg/catalog`, `POST /api/encounter/{id}/ecg`.

• **Acceptance:**

• `tests/v7/test_ecg_catalog_lists_eleven_built_in_waveforms.py`

• `tests/v7/test_ecg_set_per_encounter_persists.py`

• Manual browser: M25 picker renders all 11 strips, switching

between them is visible.

• **Blocks:** M25, M27.

• **Blocked by:** M1.2.

• **Est. effort:** 2 days (mostly waveform math + renderer).

MODULE M25 — Per-Patient Console rich features

Goal: Light up M22's scaffold with the actual instructor controls for managing a simulated patient.

• **Files touched:**

• `portal/templates/encounter_console.html` (sections added)

• `portal/static/encounter_console.{js,css}` (sections added)

• `portal/server.py` (helper functions; no new public routes

beyond what M23/M24 already added).

• **UI sections:**

1. **Live telemetry preview** — small strip showing HR/SBP/SpO2/RR/Temp using M23 snapshot, polled at 1 s. 2. **ECG strip** — uses `ecg_strip.js` to render the encounter's current waveform. Selector dropdown drives `POST /api/encounter/{id}/ecg`. 3. **Device list** — read-only list of devices bound to this encounter (uses M1.5's `list_kinds()`). 4. **Telemetry override sliders** — force HR / SBP / DBP / SpO2 / RR / Temp. Each toggleable; OFF = derived from chart events. 5. **Quick scene inject** — reuses M5's scene injector dialog, pre-targeted to this encounter.

• **Acceptance:**

• Manual browser: open Per-Patient Console for a 2-bed room → all

five sections render → ECG selector flips waveform → override slider changes the telemetry strip value. - `tests/v7/test_encounter_console_renders_all_sections.py` (smoke).

• **Blocks:** none directly; M27 mirrors its ECG strip use.

• **Blocked by:** M22, M23, M24.

• **Est. effort:** 2.5 days.

MODULE M26 — Alarm bus (`portal/alarms.py`)

Goal: A unified read of all active alarms across a room. Reads from three sources:

1. **Device events** — `alarm.injected` rows from `device_event` (existing v6 path); cleared by `alarm.cleared`. 2. **Scene events** — `instructor.trigger` rows whose payload carries `level: "alarm"` (M1.4 tag). Examples: code.blue compound marker, future fire-alarm scene. 3. **Future device stubs** — call bell, bed alarm, code blue button, fire alarm (added in M29).

Returns a per-room list ordered by severity then ts, with: alarm id, source ('device' / 'scene' / 'bedside_button'), kind, encounter id, ts, severity ('info' / 'warning' / 'critical'), cleared_at, target_label.

• **New files:**

• `portal/alarms.py` — `active_alarms(room_id) -> list[dict]`,

`clear_alarm(alarm_id) -> dict`.

• **Files touched:**

• `portal/server.py` — `GET /api/room/alarms`, `POST /api/alarm/{id}/clear`.

• **Acceptance:**

• `tests/v7/test_alarms_from_pump_appear_in_room_alarms.py`

• `tests/v7/test_alarms_from_code_blue_scene_appear.py`

• `tests/v7/test_clear_alarm_removes_it_from_active_list.py`

• **Blocks:** M27 (the nurse-station page polls this).

• **Blocked by:** M1.4.

• **Est. effort:** 2 days.

MODULE M27 — Nursing Station student role (`/portal/students/nurse_station`)

Goal: The in-sim supervisor surface. A student picks "Nursing Station" from the join page's role selector and lands on a multi-patient overview showing telemetry strips + device status + the live alarm board across every encounter in the room.

• **New files:**

• `portal/templates/nurse_station.html` — public (no operator vault).

• `portal/static/nurse_station.{js,css}`.

• **Files touched:**

• `portal/templates/student_join.html` — adds a role picker

("Bedside" / "Nursing Station") on Step 2. - `portal/static/student_join.js` — branches the POST: bedside → `/portal/students/register` (M9 existing), nurse_station → `/portal/students/register_nurse` (new). - `portal/server.py` — new `POST /portal/students/register_nurse` (creates Student with role='nurse_station', no encounter assignment) and `GET /portal/students/nurse_station?sid=...` page render.

• **Per-bed card on the nurse-station page contains:**

- Bed label + patient persona name + state badge.
- Mini ECG strip (`ecg_strip.js`).
- Telemetry strip (HR / BP / SpO2 / RR / Temp) — uses M23.
- Device pill row (color-coded by state — running / alarming / off).
- Tap-to-open intercom (M28) — placeholder button until M28 lands.
- **Acceptance:**
- `tests/v7/test_nurse_station_role_register_creates_role_row.py`
- `tests/v7/test_nurse_station_page_renders_all_encounters.py`
- Manual browser: join flow lets a student pick the nurse role →

lands on `/portal/students/nurse_station?sid=...` → sees the 2 beds with telemetry + ECG strips updating.

- **Blocks:** M28 (intercom is invoked from this page).
- **Blocked by:** M1.2 (Student.role), M23, M24, M26.
- **Est. effort:** 3 days.

MODULE M28 — Intercom (bi-directional voice)

Goal: Let the Nursing Station student talk to a specific bed and hear / read the response. The MVP cut uses the simpler path (no WebRTC): nurse-station sends a short text prompt → server synthesizes it through ElevenLabs with the encounter's *staff* voice (or a generic "Hospital Communications" voice if no staff persona is bound) → bedside chat station auto-plays it as if it were a character turn. Bedside replies via chat as normal.

A v7.1 candidate is full WebRTC audio (true mic-to-mic). Out of scope here; the data path remains the same (a `comm.intercom` chart event captures the exchange for the debrief).

- **New files:**
- `portal/intercom.py` — server-side handler for the synthesis +

chart-event recording. - `portal/static/intercom.js` — nurse-station-side widget.

- **Files touched:**
- `portal/templates/nurse_station.html` — intercom dialog.
- `portal/server.py` — `POST /api/intercom/{encounter_id}/page`.
- `portal/static/station.js` — bedside auto-plays intercom audio.
- **Acceptance:**
- `tests/v7/test_intercom_page_records_comm_event.py`
- `tests/v7/test_intercom_uses_staff_persona_voice_when_available.py`
- Manual browser: nurse station sends "Patient in Bed 2 reports

pain 8/10, need orders" → audio plays on Bed 2 station; chat log shows the turn.

- **Blocks:** none.
- **Blocked by:** M27.

• **Est. effort:** 3 days (down from a full WebRTC's 5+ — the v7.1 WebRTC upgrade is tracked as a separate candidate).

MODULE M29 — Future-device stubs (call bell, bed alarm, code blue button, fire alarm)

Goal: Add four new device kinds to the v6 device subsystem so students can press a button that surfaces as an alarm on the M26 bus. Each device is a single-button HTML page that the student joins via QR (matching the pump / cabinet device-station pattern).

• **New files (per device kind, ~50 lines each):**

• `portal/devices/call_bell/spec.json`, `engine.py`, `skin.svg`

• `portal/devices/bed_alarm/spec.json`, `engine.py`, `skin.svg`

• `portal/devices/code_blue_button/spec.json`, `engine.py`, `skin.svg`

• `portal/devices/fire_alarm/spec.json`, `engine.py`, `skin.svg`

• **Files touched:**

• `portal/devices/registry.py` — register the new kinds.

• **Acceptance:**

• Each device joins a room (existing v6 device-routes coverage).

• Pressing the button emits a `device_event` of type alarm.injected with kind: <device_kind>. - The M26 alarm bus surfaces the alarm at the nurse-station page. - `tests/v7/test_call_bell_press_appears_on_nurse_station.py` - `tests/v7/test_fire_alarm_press_appears_critical.py`

• **Blocks:** none.

• **Blocked by:** M26.

• **Est. effort:** 2 days.

3. Effort summary

Section	Modules	Days
1. Existing-module touch-ups	1.1–1.5	1.5
2.1 Per-Patient Console scaffold	M22	1.5
2.2 Telemetry simulation	M23	2.0
2.3 ECG waveform library	M24	2.0
2.4 Per-Patient Console rich features	M25	2.5
2.5 Alarm bus	M26	2.0
2.6 Nursing Station student role	M27	3.0
2.7 Intercom (no-WebRTC v1)	M28	3.0
2.8 Future-device stubs	M29	2.0
Total	**+8 modules**	**19.5**

19.5 engineer-days ≈ **4 calendar weeks for one engineer**, **2.5 weeks for two engineers running independent phases** (M22+M23+M24 can run in parallel after the 1.x touch-ups; M26+M27 once those land; M28+M29 close out).

The total v7 build (Phase 0–6 MVP at 41 days + Phase 7 at 19.5 days + Phases 8–15 at the remaining 20.5 days) is **~81 engineer-days**. Still inside the 16-week calendar window from the original P6 plan with comfortable buffer.

4. Dependency graph (post-MVP, Phase 7)

```
flowchart TD
  classDef done fill:#d4f7d4,stroke:#2a8a2a
  classDef plan fill:#fff8d4,stroke:#a08a2a

  M10[M10 MVP GATE ✓]:::done

  Touch11[1.1 Rename]:::plan
  Touch12[1.2 Student.role]:::plan
  Touch13[1.3 Drill-in route]:::plan
  Touch14[1.4 Scene level tag]:::plan
  Touch15[1.5 Device kinds API]:::plan

  M22[M22 Per-Patient scaffold]:::plan
  M23[M23 Telemetry engine]:::plan
  M24[M24 ECG library]:::plan
  M25[M25 Per-Patient rich features]:::plan
  M26[M26 Alarm bus]:::plan
  M27[M27 Nurse Station student role]:::plan
  M28[M28 Intercom]:::plan
  M29[M29 Future-device stubs]:::plan

  M10 --> Touch11 & Touch12 & Touch13 & Touch14 & Touch15
  Touch13 --> M22
  Touch12 --> M23 & M27
  Touch12 --> M24
  Touch14 --> M26
  Touch15 --> M22

  M22 --> M25
  M23 --> M25 & M27
  M24 --> M25 & M27
  M26 --> M27
  M27 --> M28
  M26 --> M29
```

5. Where this slots into the existing 22-module plan

The MVP plan (M0–M10) is complete. The remaining v6 plan modules (M11–M21) still apply — they're the "Full Build" features (Activities, dual chart mode, cohort debrief, WebSocket, cost caps, observer, capacity, Playwright, LAN release gate).

Phase 7 (M22–M29) is **additive** to that plan. Three slotting options:

1. **After M21 (release gate)** — Phase 7 begins after the formal LAN release of the v6-plan scope. Operator demos the MVP + full-build features, then Phase 7 lands as v7.1. 2. **Before M21, after M15 (cohort debrief)** — slot Phase 7 in before WebSocket / cost caps / observer / capacity / LAN release so the release gate covers

Phase 7 too. Adds ~4 weeks before LAN release. Most ambitious. 3. **In parallel with M11–M19** — Phase 7 modules with no overlap on the v6-plan modules can run on a second engineer independently. M22/M23/M24/M26 touch new files entirely; M27/M28 share `student_join.html` (M9) and `station.js` (v3 chat) so they'd serialize on those files.

Recommendation: Option 1 (after M21). Reasons:

• The MVP gate is freshly verified — Phase 7 builds on a stable

base, not a moving target.

• Operator can demo + collect classroom feedback (the original M10

pause-here intent) including feedback on whether the nursing-station role is required at the scale they actually deploy.

• Phase 7 introduces no v6-regression risk because nothing in

Phase 7 modifies v6's contract; the M10 byte-for-byte test stays green.

• WebSocket transport (M16) lands before Phase 7 — the

nursing-station telemetry strip would feel laggy on HTTP poll alone, and M16's per-room channel push is exactly the right vehicle.

6. Pause/resume contract

Phase 7 modules follow the same `BUILD_STATE.md` + per-module guide contract as M0–M21:

• Each module gets its own `M<NN>_<slug>.md` in

`docs/module_guides/` with Purpose / Files / Acceptance / Change list / Open questions.

• Rendered to PDF via `render_pdfs.py`.

• Phase-table row in `BUILD_STATE.md` flips to DONE only when

acceptance tests are green AND the manual browser flow (where applicable) has been verified.

• `CONTINUATION.md`'s seven-step resume protocol works unchanged.

7. Out of scope (explicit)

These have been mentioned in the source clarification but are NOT in Phase 7:

• **Real bi-directional WebRTC audio** between Nursing Station and

bedside. M28 ships a one-way TTS-to-bedside path with text-back. WebRTC is a v7.1 candidate; the data contract (`comm.intercom` chart event) is forward-compatible.

• **Autonomous physiological model.** Telemetry (M23) interpolates

around the latest `vitals.record`; it does not drive vitals forward in time on its own. The physiology engine is the v7.2 candidate (referenced in P6 §10).

• **Telemetry alarms on out-of-range vitals.** M26 surfaces alarms

from device events and scene events. A future module would let the instructor configure threshold alarms ("HR > 130 triggers tachy alarm") that automatically fire on M23's telemetry — that needs the threshold-editor UI plus the alarm-bus interface, ~1.5 days of work, deferred.

• **Voice-recognition return path** from bedside. M28 sends voice

one-way (nurse → bed). A bedside student's microphone reply goes through the existing v6 student chat path as text.

• **Recording / archiving** of intercom audio. The `comm.intercom`

event stores the TEXT and a reference to the synthesized voice ID; the audio itself is regenerated on demand (no PHI risk since every persona is synthetic).

8. Acceptance for the plan itself

This document is acceptable when:

• ✓ Naming clarification documented (instructor surface vs. in-sim role).

• ✓ Each new module has Purpose / Files / Acceptance / Effort.

• ✓ Dependency graph shows where M22+ slots into M0–M21.

• ✓ Effort summary fits the v6-plan's calendar budget (~16 weeks

total).

• ✓ Pause/resume protocol carries forward unchanged.

• ✓ Out-of-scope items explicitly enumerated.

The next step is operator review. On approval, M22+ start at the "recommend after M21" cadence above; in the interim the plan document lives in `docs/module_guides/PHASE7_PLAN_*` and is referenced from `BUILD_STATE.md`.